

A Forward-Horizon Obstruction for the Current Merge Table

Collatz proof project — September 5, 2026

The current table contains 360 verified rules of the form $n_0 + AQ \mapsto x_0 + BQ$, for natural Q , with smaller positive outputs and exact merging paths. This note studies only that particular table. Write $T(n) = n/2$ for even n and $(3n+1)/2$ for odd n .

Theorem 1 (Finite-table congruence classification; Lean checked). *Put $L = 2^{18}3^{11} = 46438023168$. For $0 \leq j \leq 11$, let c_j be the unique residue modulo L satisfying*

$$c_j \equiv -1 \pmod{2^{18}}, \quad c_j \equiv 3^j 2^{-j} - 1 \pmod{3^{11}},$$

where 2^{-j} denotes modular inversion. If a listed rule applies to $n \equiv c_j \pmod{L}$, it must be the inverse odd rule

$$2 + 3Q \mapsto 1 + 2Q.$$

No listed rule applies when $n \equiv c_0 \pmod{L}$.

Proof. Every rule step A divides L . The finite check verifies, for every non-inverse-odd rule and each of the twelve classes, $c_j \not\equiv n_0 \pmod{A}$. Any application would contradict this check. The remaining rule requires residue two modulo three, whereas c_0 is divisible by three.

The exact representatives are

42120773631	16743137279	25114705919	14453047295
21679570943	9300344831	13950517247	44144787455
42998169599	41278242815	15479341055	46438023167.

Lean checks the exclusions for these explicit representatives by ordinary `decide`, and proves their extension to every natural quotient. The Chinese-remainder interpretation is reproduced by the exact Python certificate; it is not an additional Lean theorem. $\square$

Theorem 2 (Arbitrary fixed horizon; written deduction). *For every natural K , there are arbitrarily large positive seeds N with noncontracting prefixes through K such that, for every $k \leq K$, every chain of the listed rules starting at $T^k(N)$ stops at or above N . Chain length is unrestricted. In particular, no such chain supplies a smaller replacement for N .*

Proof. Choose any positive solution of

$$N \equiv -1 \pmod{2^{K+18}}, \quad N \equiv 0 \pmod{3^{11}}.$$

The moduli are coprime, so solutions form an unbounded arithmetic progression. The first K steps are odd, and

$$y_k := T^k(N) = \frac{3^k(N+1)}{2^k} - 1 \quad (0 \leq k \leq K).$$

Consequently $y_k \equiv -1 \pmod{2^{18}}$ and $y_k \equiv 3^k 2^{-k} - 1 \pmod{3^{11}}$. For $k \geq 11$ the latter residue is stationary at -1 . Thus y_k belongs to class $c_{\min(k,11)}$.

For $k > 0$, the only possible rule is the inverse odd rule, and its output is exactly y_{k-1} , since $3y_{k-1} + 1 = 2y_k$. At $k = 0$ there is no listed rule. Induction therefore restricts every chain to $y_k, y_{k-1}, \dots, N$. All these seeds are at least N , and the prefix coefficients are $(3/2)^i \geq 1$. Under deficit transport the initial deficit $(3/2)^k$ reaches one only on returning to N . $\square$

Formal status and reproduction. The first theorem's classification of the explicit residues is kernel checked in `Collatz.MergeRuleObstruction`; both public theorems enter the axiom audit. The arbitrary-horizon deduction is a written proof, not yet formalized in Lean. `analysis/merge_rule_obstruction.py` reconstructs the residues and the CRT seeds; independent tests replay horizons through 128 and multiple progression quotients. Those examples are checks of the construction, not the reason its quantifier over K holds.

Implication for the proof program. No uniformly bounded forward horizon combined with arbitrarily long chains from this fixed table covers all finite-prefix candidates. This does not refute coverage with a seed-dependent horizon, a larger rule family, or another strategy. The constructed seeds are not asserted to be fully noncontracting or divergent. Collatz remains unproved; no mathematical priority claim is made.